

1. Split your app into at least two routers: `routers/users.py` and `routers/auth.py`. Register both in `main.py` with appropriate prefixes and tags.

FastAPI 0.1.0 OAS 3.1
/openapi.json

Users

GET	/users/	Get Users
POST	/users/	Create User
Authentication		
POST	/auth/	Login

create_user_users__post

Users

GET /users/ Get Users

Parameters

No parameters

Execute Clear

Responses

Curl

curl -X 'GET' \n'http://127.0.0.1:8000/users/' \n-H 'accept: application/json'

Request URL

http://127.0.0.1:8000/users/

Server response

Code Details

200

Response body

{\n "message": "All users"\n}

Response headers

content-length: 23\ncontent-type: application/json\ndate: Mon, 13 Jul 2025 18:30:58 GMT\nserver: uvicorn

Response

Code Description Links

200 Successful Response No links

Media type

application/json

Controls Accept header.

Example Value Schema

"string"

POST /users/ Create User

Parameters

No parameters

Execute Clear

2. Add a middleware that logs every request method, path, and response status code to the console.'

```
main.py × users.py auth.py
main.py > log_request
3
4 app=FastAPI()
5 app.include_router(users.router)
6 app.include_router(auth.router)
7
8 from fastapi import Request
9 @app.middleware('http')
10 async def log_request(request: Request, call_next):
11     response=await call_next(request)
12     print(
13         request.method,
14         request.url.path,
15         response.status_code
16     )
17     return response
18
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
(venv) (base) PS C:\Users\Alina\Desktop\Internship-Artificizen\Week 2\Day 5> uvicorn main:app --reload
INFO:     Waiting for application startup.
INFO:     Application startup complete.
GET /docs 200
INFO:     127.0.0.1:59544 - "GET /docs HTTP/1.1" 200 OK
GET /openapi.json 200
INFO:     127.0.0.1:59544 - "GET /openapi.json HTTP/1.1" 200 OK
GET /users/ 200
INFO:     127.0.0.1:50654 - "GET /users/ HTTP/1.1" 200 OK
POST /auth/ 200
INFO:     127.0.0.1:51705 - "POST /auth/ HTTP/1.1" 200 OK
POST /auth/ 200
INFO:     127.0.0.1:51705 - "POST /auth/ HTTP/1.1" 200 OK
POST /users/ 200
INFO:     127.0.0.1:58331 - "POST /users/ HTTP/1.1" 200 OK
```

3. Add CORSMiddleware configured to allow requests from `http://localhost:3000` only.

```
from fastapi.middleware.cors import CORSMiddleware
app.add_middleware(CORSMiddleware, allow_origins=[
    "http://localhost:3000"
],
    allow_credentials=True,
    allow_methods=["*"],
    allow_headers=["*"]
)
```

4. Write a global exception handler for `HTTPException` that always returns JSON in the shape `{"error": true, "detail": "...", "status": 404}` instead of FastAPI's default.

GET /test-error Test Error

Parameters

No parameters

Execute Clear

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/test-error' \
  -H 'accept: application/json'
```

Request URL

http://127.0.0.1:8000/test-error

Server response

Code	Details
404	Error: Not Found

Response body

```
{
  "error": true,
  "detail": "User not found",
  "status": 404
}
```

Response headers

```
content-length: 53
content-type: application/json
date: Mon, 13 Jul 2025 12:46:18 GMT
server: unicorn
```

Responses

Code	Description	Links
200	Successful Response	No links

Media type

application/json

Controls Accept header:

Example Value : Schema

```
"string"
```

5. Add a background task to the POST /users route that “sends a welcome email” (just print it) without blocking the response.

```
main.py  users.py  test_main.py  auth.py
routers > users.py > create_user
6 )
7
8 def send_welcome_email():
9     print("Welcome email sent!")
10
11 @router.get("/")
12 def get_users():
13     return {"message": "All users"}
14
15 @router.post("/")
16 def create_user(background_tasks: BackgroundTasks):
17
18     background_tasks.add_task(send_welcome_email)
19
20     return {
21         "message": "User Created Successfully"
22     }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
(venv) (base) PS C:\Users\Alina\Desktop\Internship-Artificizen\Week 2\Day 5> uvicorn main:app --reload
INFO:      Finished server process [30320]
INFO:      Started server process [32700]
INFO:      Waiting for application startup.
INFO:      Application startup complete.
GET /docs 200
INFO:      127.0.0.1:57145 - "GET /docs HTTP/1.1" 200 OK
GET /openapi.json 200
INFO:      127.0.0.1:57145 - "GET /openapi.json HTTP/1.1" 200 OK
POST /users/ 200
INFO:      127.0.0.1:57145 - "POST /users/ HTTP/1.1" 200 OK
Welcome email sent!
POST /users/ 200
INFO:      127.0.0.1:57145 - "POST /users/ HTTP/1.1" 200 OK
Welcome email sent!
```

6. Write at least three pytest tests using TestClient: one for a successful user creation, one for a duplicate email conflict, and one for a protected route returning 401 without a token.

```
(venv) (base) PS C:\Users\Alina\Desktop\Internship-Artificizen\Week 2\Day 5> python -m pytest
===== test session starts =====
platform win32 -- Python 3.13.5, pytest-9.1.1, pluggy-1.6.0
rootdir: C:\Users\Alina\Desktop\Internship-Artificizen\Week 2\Day 5
plugins: anyio-4.14.2
collected 3 items

test\test_main.py ... [100%]
```